Exploring shapely

In [1…
```
import shapely

p = shapely.Point(1,2)
```

In [2…
```
p
```

Out[2…

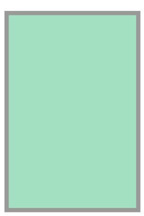
That isn't a placeholder, the point is being displayed on the screen.

In [3…
```
p.xy
```

Out[3…
```
(array('d', [1.0]), array('d', [2.0]))
```

In [4…
```
poly = shapely.Polygon([(0,0),(2,0),(2,3),(0,3),(0,0)])
poly
```

Out[4…



In [5…
```
poly.contains(p)
```

Out[5…
```
True
```

There are various points that can stand in for the whole object: `centroid` and `representative_point`.

In [6…
```
poly.centroid
```

Out[6…

In [7…
```
print(poly.centroid)
```

```
POINT (1 1.5)
```

In [8…
```
type(poly.centroid)
```

Out[8…
```
shapely.geometry.point.Point
```

In [9…
```
rp = poly.representative_point()
print(rp)
```

```
POINT (1 1.5)
```

In [1…
```
type(rp)
```

Out[1…
```
shapely.geometry.point.Point
```

In [1…
```
q = shapely.Point(4,5)
q.distance(p)
```

Out[1…
```
4.242640687119285
```

A point has an attribute `coords` whose type is:

In [1…
```
type(p.coords)
```

Out[1…
```
shapely.coords.CoordinateSequence
```

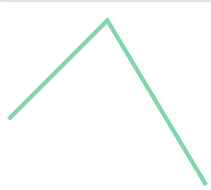
An easy way to get at a `CoordinateSequence` is to turn it into a `list`:

In [1…
```
list(p.coords)
```

Out[1…
```
[(1.0, 2.0)]
```

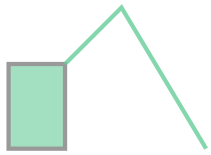
In [1…
```
r = shapely.Point(7,0)
LS = shapely.LineString([p,q,r])
LS
```

Out[1…



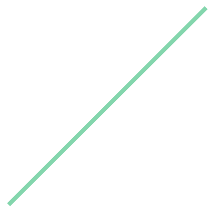
In [1…
```
LS.union(poly)
```

Out[1…



In [1…
```
LS.intersection(poly)
```

Out[1…



In [1…
```
type(LS)
```

Out[1…
```
shapely.geometry.linestring.LineString
```

In [1…
```
type(LS.coords)
```

Out[1…
```
shapely.coords.CoordinateSequence
```

In [1…
```
LS.coords.xy
```

Out[1…
```
(array('d', [1.0, 4.0, 7.0]), array('d', [2.0, 5.0, 0.
0]))
```

In [2…
```
list(LS.coords)
```

Out[2…
```
[(1.0, 2.0), (4.0, 5.0), (7.0, 0.0)]
```

There is a much more complicated way to dig down into a `CoordinateSequence`:

In [2…
```
LS.coords[:][0]
```

Out[2…
```
(1.0, 2.0)
```

The brackets colon notation in Python means to make a copy of a list. Here it converts a `CoordinateSequence` object into a `list`.

In [2…
```
type(LS.coords[:])
```

Out[2…
```
list
```

Of course, a list can have more than one element. Hence the `[0]`.

In [2…

```
print(poly)
```

```
POLYGON ((0 0, 2 0, 2 3, 0 3, 0 0))
```

Notice the double parentheses. The reason for this is that it is allowed for a polygon to have holes. (For example, a state might contain lakes).

In [2…

```
border = [(-180, 90), (-180, -90), (180, -90), (180, 90)]
holes = [[(-170, 80), (-170, -80), (170, -80), (170, 80)]
frame = shapely.Polygon(shell=border, holes=holes)
print(frame)
```

```
POLYGON ((-180 90, -180 -90, 180 -90, 180 90, -180 90), (
-170 80, -170 -80, 170 -80, 170 80, -170 80))
```

In [2…

```
frame
```

Out[2…

In [2…

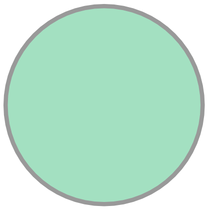
```
print(f"Polygon centroid: {frame.centroid}")
print(f"Polygon Area: {frame.area}")
print(f"Polygon Bounding Box: {frame.bounds}")
print(f"Polygon Exterior: {frame.exterior}")
print(f"Polygon Exterior Length: {frame.exterior.length}"
```

```
Polygon centroid: POINT (0 0)
Polygon Area: 10400.0
Polygon Bounding Box: (-180.0, -90.0, 180.0, 90.0)
Polygon Exterior: LINEARRING (-180 90, -180 -90, 180 -90,
180 90, -180 90)
Polygon Exterior Length: 1080.0
```

In [2…

```
# Circle (using a buffer around a point)
point = shapely.Point((0,0))
point.buffer(1)
```

Out[2…



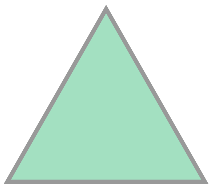
In [2…

```
import geopandas as gpd
import pandas as pd
from shapely import Polygon, Point
from shapely import plotting
```

In [2…

```
pg = Polygon([(0,0),(4,0),(2,3.5),(0,0)])
sq = Polygon([(2,2),(3,2),(3,4),(2,4),(2,2)])
pg
```

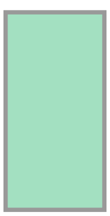
Out[2…



In [3…

```
sq
```

Out[3…



In [3…

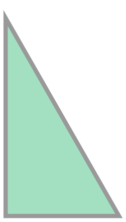
```
pg.intersects(sq)
```

Out[3… True

In [3…

```
pg.intersection(sq)
```

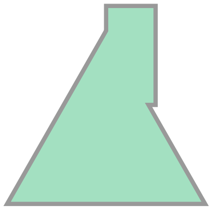
Out[3…



In [3…

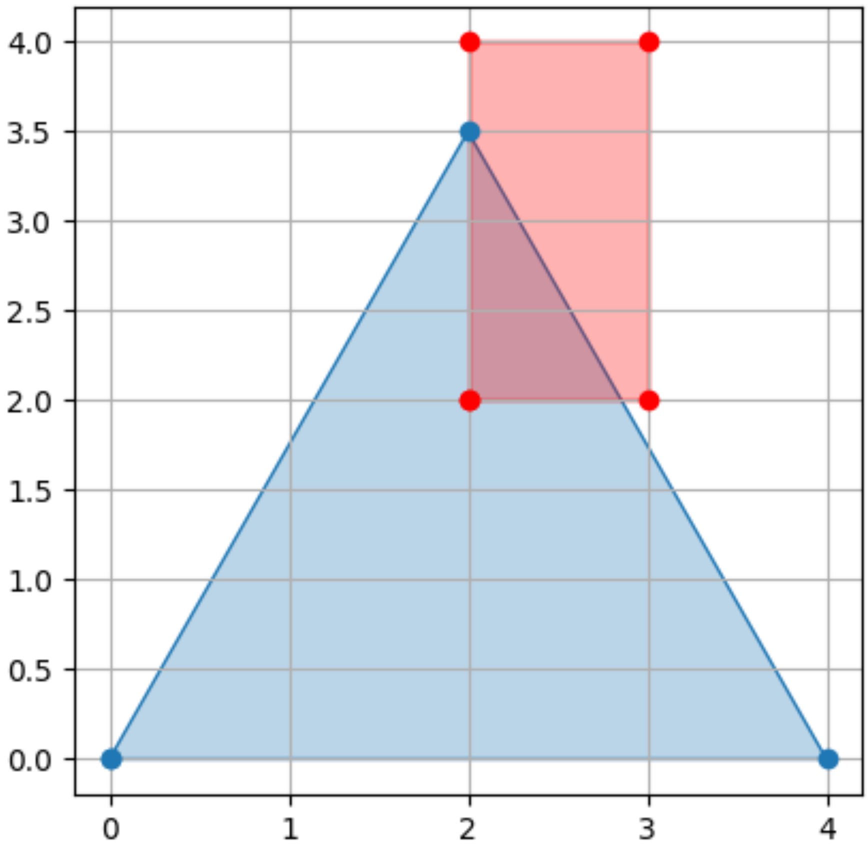
```
pg.union(sq)
```

Out [3…



In [3…
```
plotting.plot_polygon(pg)
plotting.plot_polygon(sq,color='r')
```

Out [3…
```
(<matplotlib.patches.PathPatch at 0x110d311d0>,
 <matplotlib.lines.Line2D at 0x110d31310>)
```



In [3…
```
pg.boundary.intersection(sq.boundary).wkt
```

Out [3…
```
'MULTIPOINT ((2 3.5), (2.857142857142857 2))'
```

In [3…
```
b = pg.boundary
type(b)
```

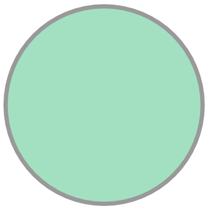
Out [3…
```
shapely.geometry.linestring.LineString
```

In [3…
```
b.coords
```

Out [3…
```
<shapely.coords.CoordinateSequence at 0x110d61b00>
```

In [3…
```
coords = list(b.coords)[2]
c = Point(coords).buffer(1)
c
```

Out[3…



In [3…
```
cent = pg.centroid
cent.y
```

Out[3…
```
1.1666666666666665
```

In [4…
```
type(cent.y)
```

Out[4…
```
float
```

In [4…
```
plotting.plot_points(c.boundary)
```

Out[4…
```
<matplotlib.lines.Line2D at 0x110e0dbd0>
```